

Using Advantech DMS-ST16 (Power Cycle Testing Device)

Purpose: This testing device can be used to detect problems during power on/off of a defective computer. This device can be used for multiple types of testing such as AT, ATX, COM, and LED mode. This guide will give you basic instructions on how to configure and set up all components to properly test computer/DUT (device under testing).

Sections:

- 1) Basic power on/off testing without software (minimal setup)
- 2) Power on/off testing with POWERUP UTILITY software & COM port.

Section 1:

Setup Requirements:

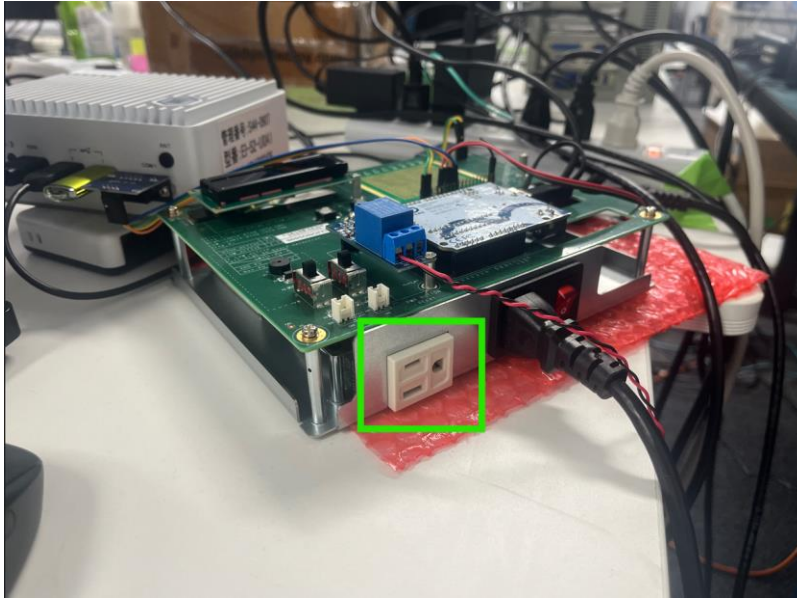
- Device Under Test (DUT) & power cable
- DMS-ST16 Power on/off testing device & power cable

Purpose:

- This testing method should be used to test if the DUT can properly turn on/off without any issues. If there is an issue, you should switch to method 2 to identify the problem. This is because method 1 lacks any way to identify the issue. This method is good for its basic testing and easy setup

Process:

1.) Plug the testing device (ST16) into a power outlet and plug the DUT's cable into the testing device's power outlet (green box in picture). Turn the power switch of the ST16 on.



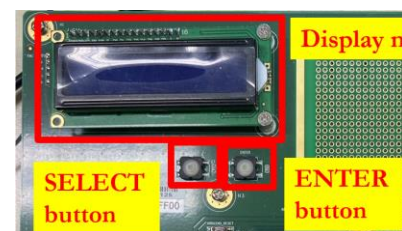
2.) For minimal setup, turn the DUT into AT powerup mode (either with a hardware jumper on the motherboard or through configuration in the BIOS). This will cause the DUT to power on directly from AC power and not with the need to push the power button.

3.) Setup the AT testing parameters using the display monitor on the ST16 and perform testing. (PPT 12,17,19,20,24,28)

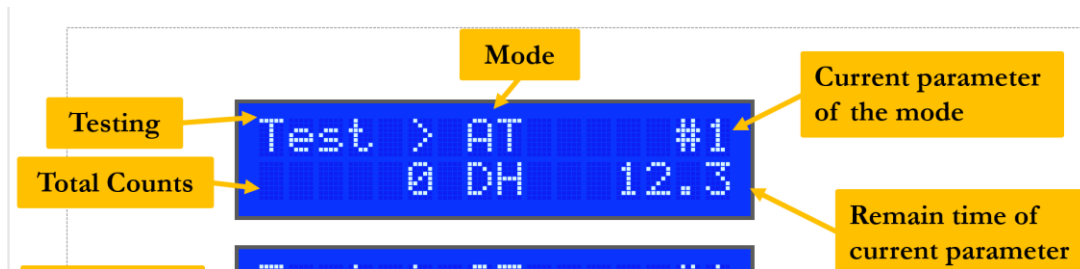
Basic controls for setup: (PPT 16,17)

4-5-2 Under Setting condition:

- SELECT : press and hold - go back to the previous menu
press once - select different items in that menu or reduce value when it requires to enter values.
- ENTER : press and hold - confirm and save the setting/value
press once - enter current menu/item or add value when it requires to enter values



- 1) Go to page 4 and disable COM Port, COM Enable, and Error Hold.
- 2) Go to page 2, choose AT mode, choose AcOffTm (time that the DUT is off during testing). Should choose at least 10 seconds to give the DUT time to properly turn off. Choose AcOnTm (time that the DUT is on during testing) should be at least 20 seconds. (PPT 19, 20, 21)
- 3) Go to page 1, choose AT mode, and start. (PPT 18)
- 4) Testing should now be going on (PPT 29)



5) To stop testing, hold the select button. (PPT 28)

Section 2:

Setup Requirements:

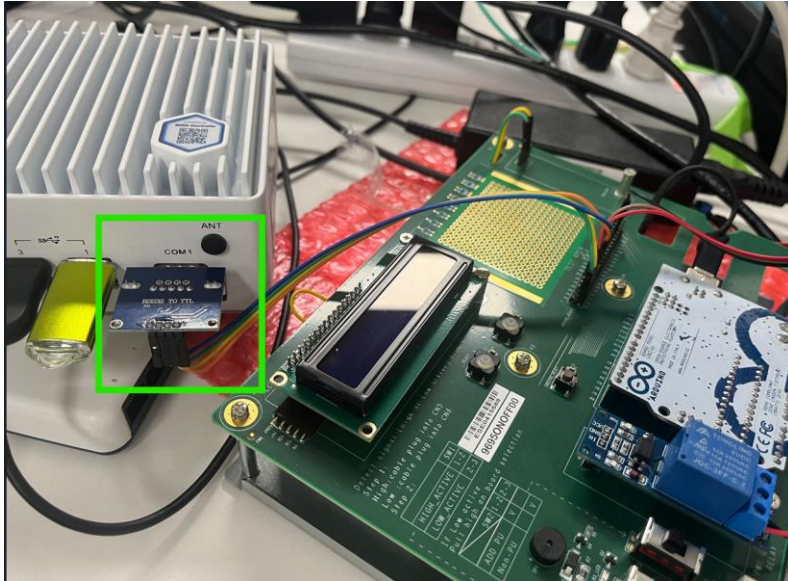
- Device Under Test (DUT) with keyboard & power cable
- USB Flash drive (FAT32 Format) with POWERUP utility v3.7 software
- DMS-ST16 testing device with COM extension & power cable

Purpose:

- This testing method should be used to identify the problem of the DUT. Using the powerup utility software provides more descriptive data and information about tests such as temperature, CPU info, etc. Using powerup also provides the opportunity of COM support. This can be used for error hold, which is a setting that stops the testing if an error occurs.

Process:

1.) Plug the testing device (ST16) into a power outlet and plug the DUT's cable into the testing device's power outlet (PPT-8). Plug the COM extension from the testing device into the COM port of your DUT (PPT-7).



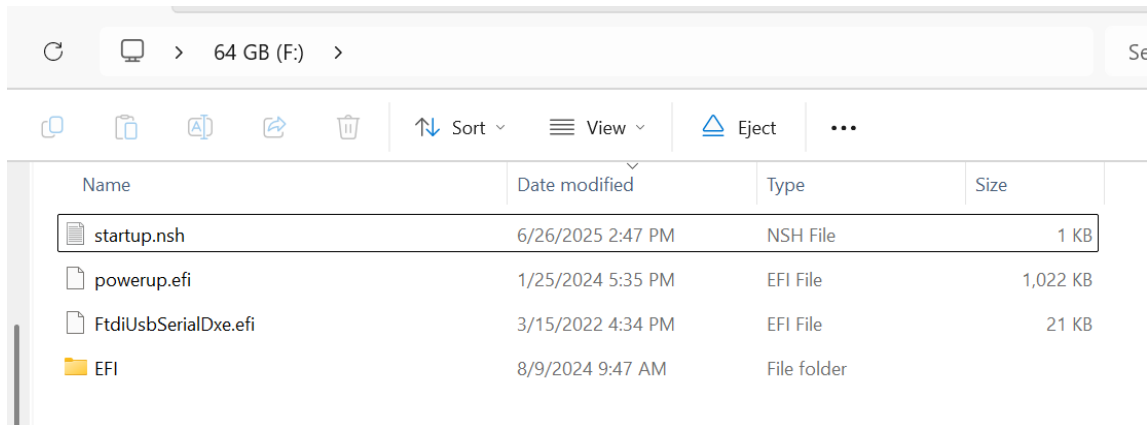
2.) Obtain a USB flash drive (and turn to FAT32 format if not already). Plug the flash drive into a system different from your DUT. Download [Powerup v3.7 download](#) and copy EFI, FtdiUsbSerialDxe.efi, and powerup.efi **INTO THE ROOT** of the flash drive.

3.) Create a startup.nsh file in the root of the flash drive folder. This will cause the UEFI shell to automatically open up the powerup utility software every time the DUT turns on.

- 1) In root file folder, click the “view” dropdown, click “show”, and click “file name extensions”
- 2) In the root folder, right click and create a “text document” named “startup.nsh”
- 3) Open the text with notepad and type in these commands. **PS: if this doesn't work, try fs0**

```
File Edit View
fs1:
powerup.efi|
```

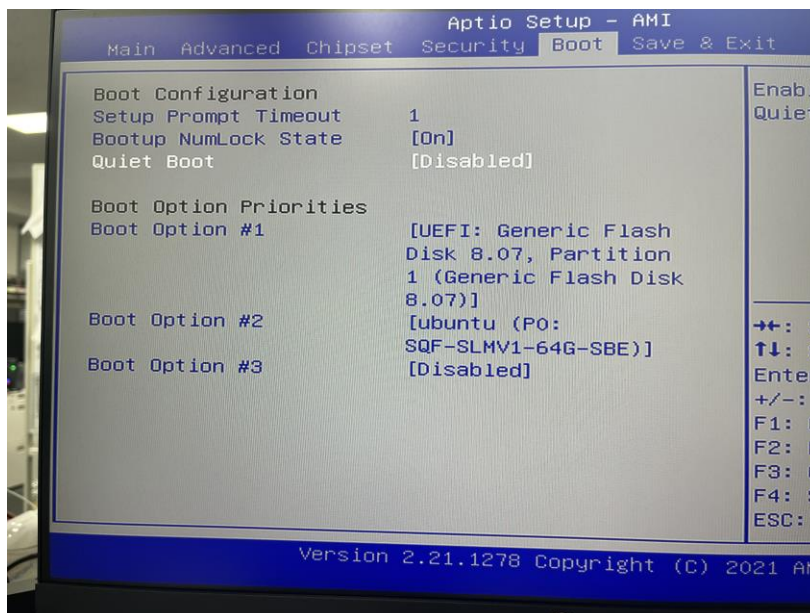
- 4) Save the file and close it. The root folder should now look like this



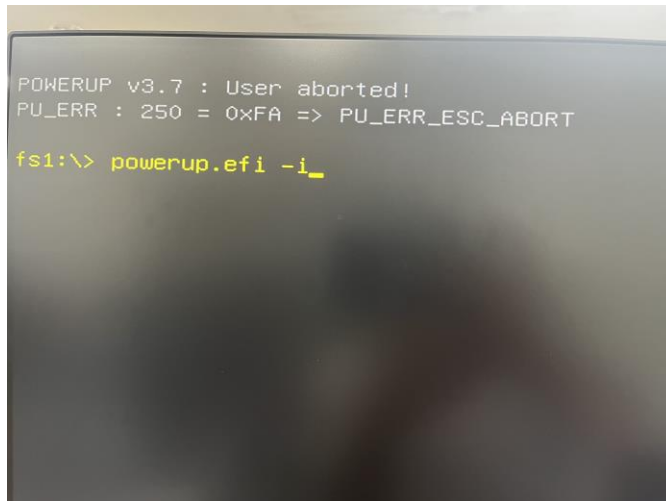
5) Press eject and unplug the flash drive from your system

4.) Plug the flash drive into your DUT and turn the power switch of the ST16 on.

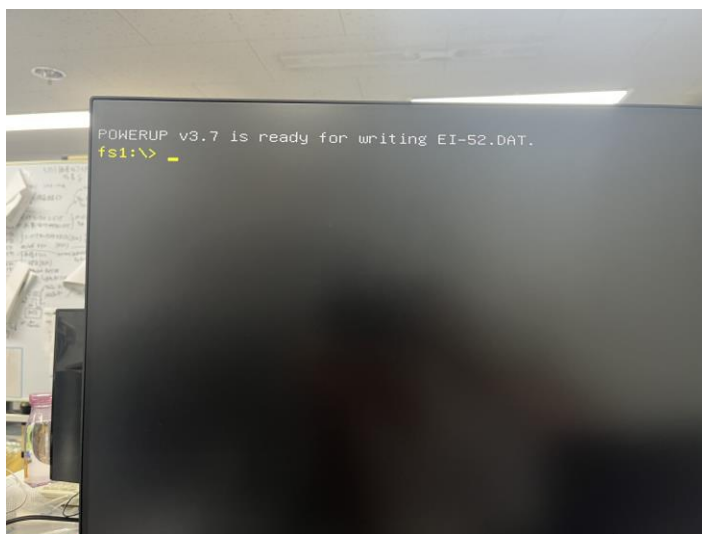
5.) In the DUT's bios, change boot option #1 to the flash drive then save and exit.



6.) When you setup the testing parameters using the powerup utility software, you don't need to do it on the ST16. To setup the parameters type in "powerup.efi -i" in the command line



7.) To use COM support, you first need to enable it by going to page 4 on the display of the ST16. After that, you can configure AT mode with COM support through the software using PPT slides 31-43. This is where you can also choose to setup error hold. Error hold pauses the testing cycle any time an error occurs. This can help you identify the root of the problem. Once done press “ok” and enter a couple times. It should then send you back to the command line where it says “POWERUP v3.7 is ready for writing “name”.DAT”. This means that the DUT is configured and ready for testing.

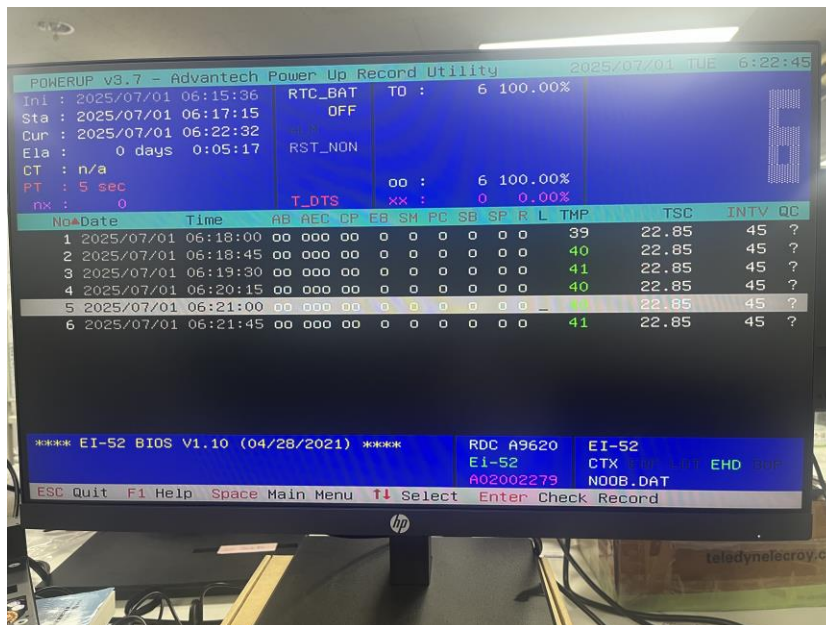


8.) Start the testing through the ST16 by going to page 1, choosing AT mode, and starting. (PPT 18)

9.) Stop the testing, hold the select button on the ST16. Next, press “ESC” and exit the powerup menu.

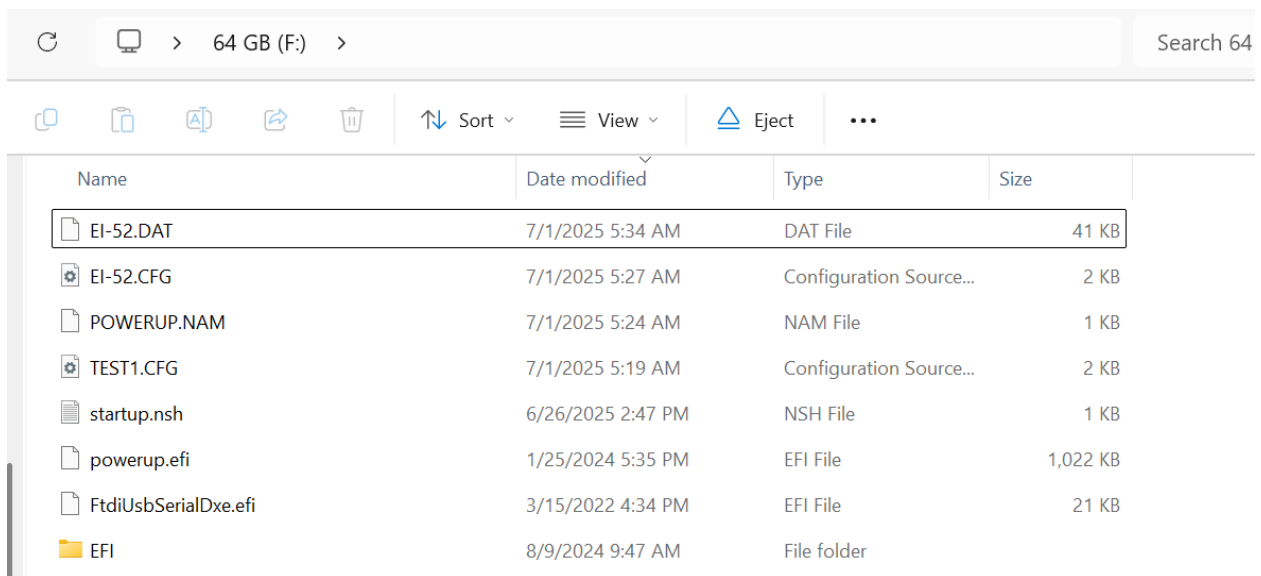
- 1) Hold the select button on the ST16
- 2) Press “ESC” and exit the powerup menu
- 3) Type “powerup.efi -v” to enter offline mode and to view the logs
- 4) If you want the data in a text report, press F8 and press F5 to view the report.

10.) You can also view the logs by typing in the “powerup.efi” command. Through the logs you can view the results and the different computer benchmarks like temperature and voltage





11.) If you want to do another test you must pull the flash drive out of the DUT and rename the “.DAT” file in the root of the folder.



64 GB (F:) > Search				
Sort ▾ View ▾ Eject ...				
Name	Date modified	Type	Size	
EI-52_OLD.DAT	7/1/2025 5:34 AM	DAT File	41 KB	
EI-52.CFG	7/1/2025 5:27 AM	Configuration Source...	2 KB	
POWERUP.NAM	7/1/2025 5:24 AM	NAM File	1 KB	
TEST1.CFG	7/1/2025 5:19 AM	Configuration Source...	2 KB	
startup.nsh	6/26/2025 2:47 PM	NSH File	1 KB	
powerup.efi	1/25/2024 5:35 PM	EFI File	1,022 KB	
FtdiUsbSerialDxe.efi	3/15/2022 4:34 PM	EFI File	21 KB	
EFI	8/9/2024 9:47 AM	File folder		